# Evidence-Weighted Federated Averaging: An Application of the Ω-Framework to Federated Learning

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**Abstract**

We apply the Ω-framework (Evidence-Weighted Multi-Agent Policy Gradient) to federated learning (FL). The mapping is precise: FL clients are agents, data heterogeneity across clients is evidence variance $V_i$, gradient compression is the lossy communication channel, and client selection is coalition formation. Replacing FedAvg's uniform aggregation with the evidence weight $w_i = V_{\min}/V_i$ yields *Evidence-Weighted Federated Averaging* (EW-FedAvg), which achieves a variance improvement factor of $\mathrm{HM}(V)/\mathrm{AM}(V) \leq 1$ over standard FedAvg. The self-knowledge bottleneck from the cooperative PG framework transfers directly: clients with highly non-IID data cannot improve their contribution to the global model through bandwidth alone — they need more *local data*. We prove communication-adjusted convergence rates, connect sparsified gradient communication to the blessing of dimensionality, and show that evidence-weighted client selection subsumes and unifies the robustness mechanisms of SCAFFOLD and FedProx.

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# 1 Introduction

Federated learning [1] trains a shared model across $K$ clients without centralizing data. The canonical algorithm, FedAvg, aggregates client updates using weights proportional to local dataset size $n_i/n$, where $n = \sum_i n_i$. This weighting is optimal when client data distributions are identical (IID). In practice, client data is heterogeneous (non-IID), and uniform-style aggregation produces a biased global model that converges slowly or diverges [2, 3].

The Ω-framework [5] was developed for multi-agent reinforcement learning, where agents with heterogeneous observation quality coordinate through lossy channels. Its core mechanism — the evidence weight $w_i = V_{\min}/V_i$ measuring each agent's information quality — transfers to federated learning with minimal modification. In this paper, we make this transfer precise and derive its consequences.

**Contributions.**

(1) A formal isomorphism between the Ω-framework and federated learning (Section 2).

(2) Evidence-Weighted FedAvg with a $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ variance improvement theorem (Section 3).

(3) Communication efficiency analysis via the self-knowledge bottleneck, with connections to sparse recovery (Section 4).

(4) Client selection as coalition formation with evidence-weighted rationality criteria (Section 5).

(5) Non-IID robustness analysis unifying SCAFFOLD and FedProx under the evidence-weighting lens (Section 6).

# 2 The Isomorphism

## 2.1 Setup: Federated Learning

Consider $K$ clients, each holding a local dataset $\mathcal{D}_i$ drawn from a (possibly distinct) distribution $P_i$ over input-label pairs $(x, y)$. The global objective is:

$$\min_{\theta \in \mathbb{R}^d} F(\theta) := \sum_{i=1}^{K} p_i \, F_i(\theta), \qquad F_i(\theta) := \mathbb{E}_{(x,y)\sim P_i}[\ell(\theta; x, y)], \tag{1}$$

where $p_i \geq 0$, $\sum_i p_i = 1$ are aggregation weights (typically $p_i = n_i/n$), and $\ell$ is the loss function.

At each communication round $t$, FedAvg [1] proceeds:

(i) Server broadcasts current model $\theta^t$ to selected clients.

(ii) Each selected client $i$ runs $\tau$ steps of SGD on $F_i$, producing a local update $\Delta_i^t = \theta_i^t - \theta^t$.

(iii) Server aggregates: $\theta^{t+1} = \theta^t + \sum_{i \in S_t} p_i \Delta_i^t$.

### 2.2 The Ω–FL Correspondence

Table 1: Precise mapping between the Ω-framework and federated learning.

| **Ω-Framework** | **Federated Learning** | **Formal Correspondence** |
|---|---|---|
| Agent $i \in \{1, \ldots, N\}$ | Client $i \in \{1, \ldots, K\}$ | $N = K$ |
| Policy $\pi_i \in \Pi_i$ | Local model $\theta_i \in \mathbb{R}^d$ | $\Pi_i = \mathbb{R}^d$ |
| Joint policy $\pi \in \Pi$ | Global model $\theta \in \mathbb{R}^d$ | $\theta = \sum_i p_i \theta_i$ |
| Policy gradient $\hat{v}_i$ | Local gradient $g_i = \nabla F_i(\theta)$ | $\hat{v}_i \leftrightarrow g_i$ |
| Evidence variance $V_i$ | Gradient variance $\sigma_i^2$ $= \mathbb{E}[\|g_i - \nabla F(\theta)\|^2]$ | $V_i = \sigma_i^2$ |
| Evidence weight $w_i = V_{\min}/V_i$ | Client quality weight | $w_i = \sigma_{\min}^2/\sigma_i^2$ |
| Lossy comm. channel | Gradient compression | Quantization, sparsification |
| Self-knowledge $\hat{\pi}_i$ | Client's local model | Drift $\|\theta_i - \theta\|$ |
| Self-knowledge loss $L_{\text{self}}^{(i)}$ | Client drift | $\mathbb{E}[\|\theta_i^t - \theta^t\|^2]$ |
| Coalition $S \subseteq \{1, \ldots, N\}$ | Selected client set $S_t$ | $S = S_t$ |
| Nash equilibrium $\pi^*$ | Global optimum $\theta^*$ | $\nabla F(\theta^*) = 0$ |
| Lyapunov $D(\pi) = \frac{1}{2}\|\pi - \pi^*\|^2$ | Suboptimality $F(\theta) - F(\theta^*)$ | Convergence measure |

**Remark 2.1** (Evidence variance = gradient heterogeneity)**.** *The evidence variance $V_i$ in the Ω-framework measures the noise-to-signal ratio of agent $i$'s gradient estimate. In FL, the natural counterpart is the* gradient divergence*:*

$$\sigma_i^2 := \mathbb{E}\big[\|\nabla F_i(\theta) - \nabla F(\theta)\|^2\big], \tag{2}$$

*which is driven by two sources: (a) stochastic noise from mini-batch sampling, and (b)* distribution heterogeneity $P_i \neq P_j$*. Under IID data, $\sigma_i^2$ reduces to mini-batch variance; under non-IID data, it is dominated by the distribution shift. This is precisely the structure the Ω-framework was built to handle.*

**Definition 2.2** (Gradient heterogeneity measures)**.** *For FL with $K$ clients:*

$$\sigma_i^2 := \mathbb{E}\big[\|\nabla F_i(\theta) - \nabla F(\theta)\|^2\big] \qquad \textit{(client-level variance)}, \tag{3}$$

$$\bar{\sigma}^2 := \sum_{i=1}^{K} p_i\, \sigma_i^2 \qquad \textit{(average variance} = \mathrm{AM}(\sigma^2)\textit{)}, \tag{4}$$

$$\sigma_{\min}^2 := \min_i \sigma_i^2 \qquad \textit{(best-informed client)}. \tag{5}$$

## 3 Evidence-Weighted Federated Averaging

### 3.1 The Aggregation Rule

Standard FedAvg aggregates with weights $p_i = n_i/n$ (proportional to data size). We replace this with the evidence-weighted aggregation:

**Definition 3.1** (Evidence-Weighted FedAvg)**.** *At communication round $t$, the server computes:*

$$\theta^{t+1} = \theta^t + \sum_{i \in S_t} \tilde{w}_i \, \Delta_i^t, \qquad \tilde{w}_i = \frac{w_i}{\sum_{j \in S_t} w_j}, \qquad w_i = \frac{\sigma_{\min}^2}{\sigma_i^2}. \tag{6}$$

Clients with lower gradient variance (higher evidence quality) receive higher aggregation weight. The normalization $\sum_i \tilde{w}_i = 1$ preserves the averaging property. When all $\sigma_i^2$ are equal, $w_i = 1$ for all $i$ and EW-FedAvg reduces to uniform averaging.

## 3.2 Variance Improvement

**Theorem 3.2** (EW-FedAvg variance improvement)**.** *Let $g_i^t = \nabla F_i(\theta^t) + \xi_i^t$ be the stochastic gradient of client $i$, with $\mathbb{E}[\xi_i^t] = 0$ and $\mathbb{E}[\|g_i^t - \nabla F(\theta^t)\|^2] = \sigma_i^2$. Then the aggregated gradient under EW-FedAvg satisfies:*

$$\mathbb{E}\Big[\Big\|\sum_{i=1}^{K} \tilde{w}_i \, g_i^t - \nabla F(\theta^t)\Big\|^2\Big] \;\le\; \frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)} \cdot \bar{\sigma}^2 \;\le\; \bar{\sigma}^2, \tag{7}$$

*where $\mathrm{HM}(\sigma^2) = K/\big(\sum_{i=1}^K 1/\sigma_i^2\big)$ and $\mathrm{AM}(\sigma^2) = (1/K)\sum_{i=1}^K \sigma_i^2$. The ratio $\mathrm{HM}(\sigma^2)/\,\mathrm{AM}(\sigma^2) \le 1$ with equality iff all $\sigma_i^2$ are identical.*

*Proof.* The aggregated gradient error is:

$$\mathbb{E}\Big[\Big\|\sum_i \tilde{w}_i g_i - \nabla F\Big\|^2\Big] = \sum_i \tilde{w}_i^2 \, \sigma_i^2 \tag{8}$$

where the cross terms vanish since gradient errors across clients are independent conditional on $\theta^t$ (clients sample from independent local datasets). The weights are $\tilde{w}_i = w_i/W$ with $W = \sum_j w_j = \sigma_{\min}^2 \sum_j 1/\sigma_j^2$, so:

$$\tilde{w}_i^2 \, \sigma_i^2 = \frac{(\sigma_{\min}^2)^2}{\sigma_i^4} \cdot \frac{\sigma_i^2}{W^2} = \frac{(\sigma_{\min}^2)^2}{\sigma_i^2 \, W^2}. \tag{9}$$

Summing:

$$\sum_i \tilde{w}_i^2 \sigma_i^2 = \frac{(\sigma_{\min}^2)^2}{W^2} \sum_i \frac{1}{\sigma_i^2} = \frac{(\sigma_{\min}^2)^2}{W^2} \cdot \frac{W}{\sigma_{\min}^2} = \frac{\sigma_{\min}^2}{W} = \frac{1}{\sum_i 1/\sigma_i^2}. \tag{10}$$

This is $\mathrm{HM}(\sigma^2)/K$. Now, uniform averaging gives $\bar{\sigma}^2/K = \mathrm{AM}(\sigma^2)/K$ (each client weighted $1/K$, contributing $\sigma_i^2/K^2$, summing to $\mathrm{AM}(\sigma^2)/K$). Therefore:

$$\frac{\text{EW-FedAvg variance}}{\text{FedAvg variance}} = \frac{\mathrm{HM}(\sigma^2)/K}{\mathrm{AM}(\sigma^2)/K} = \frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)} \le 1, \tag{11}$$

where the final inequality is the HM-AM inequality. Equality holds iff all $\sigma_i^2$ are equal. □

**Remark 3.3** (Magnitude of improvement)**.** *If $K = 2$ clients have $\sigma_1^2 = 1$ and $\sigma_2^2 = 100$, then:*

$$\frac{\mathrm{HM}}{\mathrm{AM}} = \frac{2/(1 + 1/100)}{(1 + 100)/2} = \frac{200/101}{101/2} = \frac{400}{10201} \approx 0.039.$$

*The variance is reduced by a factor of $\sim 25\times$. Highly heterogeneous settings produce the largest gains — precisely the regime where FedAvg struggles most.*

### 3.3 Convergence of EW-FedAvg

**Theorem 3.4** (Convergence rate of EW-FedAvg)**.** *Assume $F$ is $L$-smooth, each $F_i$ is $\mu$-strongly convex, and $\mathbb{E}[\|g_i - \nabla F_i\|^2] \le \sigma_{sgd}^2$ (mini-batch noise). After $T$ communication rounds with $\tau$ local SGD steps each and step size $\eta = 1/(4L\tau)$:*

$$\mathbb{E}[F(\theta^T)] - F^* \;\le\; \underbrace{\left(1 - \frac{\mu}{4L}\right)^T}_{\textit{linear contraction}} \cdot (F(\theta^0) - F^*) \;+\; \frac{8L}{\mu^2}\left(\frac{\sigma_{sgd}^2}{K\tau} + \tau\,\frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)}\,\bar{\sigma}^2\right). \tag{12}$$

*Compare FedAvg [2], where the second floor term is $\tau\,\bar{\sigma}^2$ (no* HM / AM *factor).*

*Proof sketch.* Follow the standard FedAvg analysis of Karimireddy et al. [2], replacing uniform aggregation weights $p_i = 1/K$ with evidence weights $\tilde{w}_i$. The $L$-smoothness descent lemma gives:

$$\mathbb{E}[F(\theta^{t+1})] \le F(\theta^t) - \frac{\eta\tau}{2}\|\nabla F(\theta^t)\|^2 + \frac{L\eta^2\tau}{2}\sigma_{\mathrm{sgd}}^2 + \frac{L\eta^2\tau^2}{2}\sum_i \tilde{w}_i^2\,\sigma_i^2.$$

The final term is bounded by $\mathrm{HM}(\sigma^2)/(K) \cdot L\eta^2\tau^2/2$ via (10). Telescoping and using strong convexity yields (12). □

### 3.4 Algorithm

**Algorithm 1:** Evidence-Weighted Federated Averaging (EW-FedAvg)

**Input:** Initial model $\theta^0$, $K$ clients, local epochs $\tau$, learning rate $\eta$, communication rounds $T$

**Output:** Trained global model $\theta^T$

```
// Server estimates client variances (one-time or periodic)
for i = 1, ..., K do
    σ̂_i^2 ← EstimateVariance(i, θ^0);
end
σ̂_min^2 ← min_i σ̂_i^2;
for t = 0, 1, ..., T − 1 do
    // Client selection (coalition formation)
    S_t ← SelectClients({w_i}_{i=1}^K);
    Broadcast θ^t to all i ∈ S_t;
    // Local training
    for each i ∈ S_t do                                   // in parallel
        θ_i^{t,0} ← θ^t;
        for k = 0, ..., τ − 1 do
            Sample mini-batch ξ_i ~ D_i;
            θ_i^{t,k+1} ← θ_i^{t,k} − η ∇F_i(θ_i^{t,k}; ξ_i);
        end
        Δ_i^t ← θ_i^{t,τ} − θ^t;
        // Optional: sparsify Δ_i^t (Section 4)
    end
    // Evidence-weighted aggregation
    w_i ← σ̂_min^2 / σ̂_i^2 for each i ∈ S_t;
    W ← Σ_{i∈S_t} w_i;
    θ^{t+1} ← θ^t + Σ_{i∈S_t} (w_i / W) Δ_i^t;
end
return θ^T;
```

# 4 Communication Efficiency and the Self-Knowledge Bottleneck

## 4.1 The Self-Knowledge Bottleneck in FL

The cooperative PG framework established that an agent's ability to communicate its policy is fundamentally limited by its self-knowledge. In FL, the analogous statement is:

**Definition 4.1** (Client drift as self-knowledge loss)**.** *After $\tau$ local SGD steps, client $i$'s drift from the global model is:*

$$\Delta^{(i)}_{drift} := \mathbb{E}\big[\|\theta_i^{t,\tau} - \theta^t\|^2\big]. \tag{13}$$

*This is the FL analogue of the self-knowledge loss $L^{(i)}_{self}$: the local model $\theta_i$ is client $i$'s "self-model" of its contribution to the global objective.*

**Proposition 4.2** (Drift–heterogeneity relationship)**.** *Under $L$-smooth local objectives with step size $\eta \le 1/L$:*

$$\Delta^{(i)}_{drift} \;\le\; \tau^2\eta^2\,\sigma_i^2 + \tau\eta^2\,\sigma_{sgd}^2. \tag{14}$$

*The first term is driven by distribution heterogeneity; the second by stochastic noise.*

*Proof.* By $L$-smoothness and the SGD recursion:

$$\theta_i^{t,k+1} - \theta^t = \theta_i^{t,k} - \theta^t - \eta\,\nabla F_i(\theta_i^{t,k};\xi_i^k). \tag{15}$$

Taking expectations and using $\mathbb{E}[\nabla F_i(\theta_i^{t,k};\xi)] = \nabla F_i(\theta_i^{t,k})$, then bounding $\|\nabla F_i(\theta^t) - \nabla F(\theta^t)\| \le \sigma_i$ and applying the standard SGD drift analysis [2] yields (14). □

**Corollary 4.3** (Self-knowledge bottleneck in FL)**.** *The useful information in client $i$'s update $\Delta_i^t$ is bounded by:*

$$I(\theta^*;\,\Delta_i^t) \;\le\; I(\theta^*;\,\mathcal{D}_i). \tag{16}$$

*Even with* infinite bandwidth *(no compression, no quantization), client $i$'s contribution to the global model is limited by the information content of its local data about the global optimum.*

*For a highly non-IID client (large $\sigma_i^2$), the local data contains limited information about $\theta^*$. No amount of communication bandwidth compensates — the client needs more* data, *not more* bandwidth.

## 4.2 Communication-Adjusted Convergence

When clients communicate compressed updates $\hat{\Delta}_i^t$ instead of exact updates $\Delta_i^t$, a channel distortion term enters:

$$L^{(i)}_{\text{channel}} := \mathbb{E}\big[\|\Delta_i^t - \hat{\Delta}_i^t\|^2\big]. \tag{17}$$

**Theorem 4.4** (Communication-adjusted convergence)**.** *Under the conditions of Theorem 3.4, with compressed communication satisfying $L^{(i)}_{channel} \le \delta_i^2\|\Delta_i^t\|^2$ for compression ratio $\delta_i \in [0,1]$, EW-FedAvg converges at rate:*

$$\mathbb{E}[F(\theta^T)] - F^* \;\le\; \Big(1 - \frac{\mu}{4L}\Big)^T (F(\theta^0) - F^*) + \frac{8L}{\mu^2}\left(\frac{\sigma_{sgd}^2}{K\tau} + \tau\,\frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)}\,\bar{\sigma}^2 + \sum_i \tilde{w}_i^2\,\delta_i^2\,\Delta^{(i)}_{drift}\right). \tag{18}$$

*The third floor term decomposes via (14) into a self-knowledge component ($\tau^2\eta^2\sigma_i^2$) and a stochastic component ($\tau\eta^2\sigma_{sgd}^2$), both modulated by the compression ratio $\delta_i$ and the evidence weight $\tilde{w}_i$.*

*Proof sketch.* The compressed update adds noise: $\hat{\Delta}_i^t = \Delta_i^t + \epsilon_i^t$ with $\mathbb{E}[\|\epsilon_i^t\|^2] \le \delta_i^2 \|\Delta_i^t\|^2$. This enters the descent lemma as an additional variance term:

$$\mathbb{E}\big[\|\textstyle\sum_i \tilde{w}_i \epsilon_i^t\|^2\big] = \textstyle\sum_i \tilde{w}_i^2 \mathbb{E}[\|\epsilon_i^t\|^2] \le \textstyle\sum_i \tilde{w}_i^2 \delta_i^2 \Delta_{\text{drift}}^{(i)}.$$

Combining with the EW-FedAvg analysis of Theorem 3.4 gives (18). □

**Remark 4.5** (Evidence-aware compression)**.** *Theorem 4.4 suggests an optimal compression strategy: allocate more bandwidth to high-evidence clients (low $\sigma_i^2$, high $\tilde{w}_i$) since their updates carry more weight. Formally, to minimize the communication floor $\sum_i \tilde{w}_i^2 \delta_i^2 \Delta_{drift}^{(i)}$ subject to a total bandwidth constraint $\sum_i R_i \le R_{total}$, the optimal allocation is water-filling:*

$$R_i^* \;\propto\; \tilde{w}_i \sqrt{\Delta_{drift}^{(i)}}. \tag{19}$$

*High-evidence clients get more bandwidth; low-evidence clients (whose updates are downweighted anyway) can be aggressively compressed.*

### 4.3 Sparsified Gradients and the Blessing of Dimensionality

The blessing of dimensionality from the Ω-framework [6] applies directly to FL gradient communication. When the global model has $d$ parameters but the effective gradient at each client is $k$-sparse (concentrates on $k \ll d$ coordinates), compressed sensing guarantees apply:

**Proposition 4.6** (Sparse gradient recovery)**.** *If client $i$'s true gradient $\nabla F_i(\theta)$ is $k_i$-sparse (has at most $k_i$ non-negligible coordinates), then top-$k_i$ sparsification achieves:*

$$L_{channel}^{(i)} \;\le\; \frac{\sigma_{sgd}^2}{n_i} \cdot \frac{k_i \log(d/k_i)}{d}, \tag{20}$$

*requiring only $O(k_i \log(d/k_i))$ communicated coordinates instead of $d$.*

The key insight: non-IID clients tend to have *sparser* gradients. A client whose data covers only a narrow slice of the input space has gradients concentrated on the model parameters relevant to that slice. This means the clients that need the most compression (high $\sigma_i^2$, heavily non-IID) are also the most compressible.

**Corollary 4.7** (Sparsity–heterogeneity duality)**.** *Gradient sparsity $k_i$ and evidence variance $\sigma_i^2$ are positively correlated across clients: $k_i = O(1/\sigma_i^2)$ when heterogeneity is driven by data support mismatch. The evidence weight $w_i$ thus simultaneously controls aggregation quality and compression efficiency.*

## 5 Client Selection as Coalition Formation

### 5.1 Coalition Rationality in FL

In the Ω-framework, coalition $S$ is rational if its members collectively outperform their sum of individual values. In FL, the analogous question is: which subset $S_t$ of clients should the server select at round $t$?

**Definition 5.1** (Marginal contribution of client $i$)**.** *The marginal contribution of client $i$ to coalition $S_t$ is:*

$$\Delta V_i(S_t) := \mathbb{E}[F(\theta_{S_t}^{t+1})] - \mathbb{E}[F(\theta_{S_t \setminus \{i\}}^{t+1})], \tag{21}$$

*where $\theta_S^{t+1}$ denotes the model obtained by aggregating updates from clients in $S$ only.*

**Definition 5.2** (Communication-adjusted contribution)**.** *Accounting for communication cost $c_i$ (bandwidth, latency, compute), client $i$ should be included in $S_t$ iff:*

$$|\Delta V_i(S_t)| > c_i. \tag{22}$$

### 5.2 Evidence-Weighted Selection

**Theorem 5.3** (Evidence weight as selection criterion)**.** *Under $L$-smoothness and the conditions of Theorem 3.2, the marginal contribution of client $i$ satisfies:*

$$|\Delta V_i(S_t)| \;\leq\; \frac{L\eta\tau}{W^2}\, w_i \left(\|\nabla F(\theta^t)\|^2 + \sigma_i^2\right), \tag{23}$$

*where $W = \sum_{j\in S_t} w_j$. The contribution is proportional to the evidence weight $w_i$: low-evidence clients (high $\sigma_i^2$, low $w_i$) contribute less to the aggregate and are naturally deprioritized.*

*Proof sketch.* The EW-FedAvg aggregated update with client $i$ included vs. excluded differs by the weighted increment $(\tilde{w}_i/W)\Delta_i^t$. By $L$-smoothness:

$$|F(\theta_{S_t}^{t+1}) - F(\theta_{S_t\setminus i}^{t+1})| \leq L\|\theta_{S_t}^{t+1} - \theta_{S_t\setminus i}^{t+1}\| = L\cdot\frac{w_i}{W}\|\Delta_i^t\|.$$

Bounding $\mathbb{E}[\|\Delta_i^t\|] \leq \eta\tau(\|\nabla F(\theta^t)\| + \sigma_i)$ and collecting terms gives (23). □

**Corollary 5.4** (Automatic client filtering)**.** *Evidence weighting provides a soft client selection mechanism without explicit selection: even if all $K$ clients participate, the evidence weights $\tilde{w}_i = w_i/W$ automatically suppress the contribution of low-quality clients. This is strictly better than hard selection (which discards information) while achieving the same robustness.*

### 5.3 Cluster-Based Coalitions

When clients can be grouped into clusters $\mathcal{C}_1, \ldots, \mathcal{C}_M$ with similar distributions ($\sigma_i^2 \approx \sigma_j^2$ for $i, j$ in the same cluster), a two-level aggregation emerges naturally:

(i) **Intra-cluster**: average within each cluster (clients are approximately IID within a cluster, so uniform weighting is near-optimal).

(ii) **Inter-cluster**: aggregate cluster-level updates using evidence weights $w_m = \sigma_{\min}^2/\bar{\sigma}_m^2$, where $\bar{\sigma}_m^2$ is cluster $m$'s average variance.

This corresponds precisely to the hierarchical coalition structure in the $\Omega$-framework: intra-cluster cooperation (local coalitions) followed by inter-cluster weighted aggregation (global coalition).

## 6 Non-IID Robustness

### 6.1 The Non-IID Problem

The central challenge in FL is data heterogeneity. When $P_i \neq P_j$, the local objectives $F_i$ have different optima $\theta_i^* \neq \theta_j^*$, and local SGD steps drive clients away from the global optimum $\theta^*$. This "client drift" is the dominant source of convergence degradation.

Existing approaches address this through:

- **SCAFFOLD** [2]: control variates $c_i$ that correct for the difference $\nabla F_i - \nabla F$ at each client.
- **FedProx** [3]: a proximal regularizer $(\mu_{\text{prox}}/2)\|\theta - \theta^t\|^2$ added to each local objective, penalizing drift from the global model.

### 6.2 Evidence Weighting as Unified Robustness Mechanism

**Proposition 6.1** (EW-FedAvg subsumes FedProx's effect)**.** *FedProx's proximal term reduces client drift:* $\Delta_{drift}^{(i),FedProx} \leq \Delta_{drift}^{(i)}/(1+\mu_{prox}/L)^2$. *EW-FedAvg achieves an analogous effect through aggregation: the contribution of drifted clients to the global update is suppressed by* $w_i = \sigma_{\min}^2/\sigma_i^2$.

*Formally, the effective drift contribution under EW-FedAvg is:*

$$\sum_i \tilde{w}_i \, \Delta_{drift}^{(i)} \;\leq\; \frac{\mathrm{HM}(\sigma^2)}{\mathrm{AM}(\sigma^2)} \cdot \bar{\Delta}_{drift}, \tag{24}$$

*where* $\bar{\Delta}_{drift} = \sum_i (1/K)\, \Delta_{drift}^{(i)}$.

*Proof.* Since $\Delta_{\mathrm{drift}}^{(i)} \leq C\,\sigma_i^2$ for some constant $C$ depending on $\eta, \tau$ (Theorem 4.2):

$$\sum_i \tilde{w}_i \, \Delta_{\mathrm{drift}}^{(i)} \leq C \sum_i \tilde{w}_i \, \sigma_i^2 = C \sum_i \frac{w_i}{W} \sigma_i^2 = C \sum_i \frac{\sigma_{\min}^2}{W} = \frac{CK\sigma_{\min}^2}{W} = \frac{C \cdot \mathrm{HM}(\sigma^2)}{1}. \tag{25}$$

Meanwhile, $\bar{\Delta}_{\mathrm{drift}} \leq C \cdot \mathrm{AM}(\sigma^2)$, giving the ratio (24). □

**Proposition 6.2** (EW-FedAvg complements SCAFFOLD)**.** *SCAFFOLD's control variates correct the bias:* $\mathbb{E}[g_i - c_i] = \nabla F(\theta)$. *This reduces* $\sigma_i^2$ *to the residual variance after control-variate correction:*

$$\sigma_i^{2,SCAFFOLD} = \mathbb{E}\big[\|g_i - c_i - \nabla F(\theta)\|^2\big] \;\leq\; \sigma_{sgd}^2. \tag{26}$$

*When* $\sigma_i^{2,SCAFFOLD} \approx \sigma_{sgd}^2$ *for all clients, evidence weights become approximately uniform:* $w_i \approx 1$. *EW-FedAvg gracefully reduces to standard averaging when heterogeneity is already controlled. The two mechanisms are complementary:*

| ***Mechanism*** | ***Target*** | ***Action*** |
|---|---|---|
| *SCAFFOLD* | *Bias* $\mathbb{E}[g_i] \neq \nabla F$ | *Corrects via control variate* |
| *EW-FedAvg* | *Variance Var*$(g_i)$ *heterogeneity* | *Reweights via* $w_i$ |
| *FedProx* | *Client drift* $\|\theta_i - \theta\|$ | *Penalizes via regularizer* |

### 6.3 Convergence Under Arbitrary Heterogeneity

**Theorem 6.3** (Heterogeneity-adaptive convergence)**.** *Under the conditions of Theorem 3.4, EW-FedAvg achieves:*

$$\mathbb{E}[F(\theta^T)] - F^* \;\leq\; \Big(1 - \frac{\mu}{4L}\Big)^T (F(\theta^0) - F^*) + \frac{8L\tau}{\mu^2} \frac{K}{\sum_{i=1}^K 1/\sigma_i^2} + \frac{8L}{\mu^2} \cdot \frac{\sigma_{sgd}^2}{K\tau}. \tag{27}$$

*In the homogeneous limit* ($\sigma_i^2 = \sigma^2$ *for all* $i$)*, this recovers FedAvg's rate with floor* $8L\tau\sigma^2/\mu^2$. *In the heterogeneous regime, the floor is determined by the* harmonic mean *of the variances — dominated by the best clients rather than the worst.*

**Remark 6.4** (Comparison with existing bounds)**.** *FedAvg's convergence floor scales with* $\mathrm{AM}(\sigma^2)$ *— the worst clients dominate. FedProx scales with* $\mathrm{AM}(\sigma^2)/(1+\mu_{prox}/L)^2$ *— reduced but still dominated by worst clients. EW-FedAvg scales with* $\mathrm{HM}(\sigma^2)$ *— dominated by the* best *clients. This is the fundamental advantage: evidence weighting leverages information quality differences rather than suffering from them.*

# 7 The Full Algorithm

Combining all components — evidence-weighted aggregation, communication efficiency, and adaptive client selection — we obtain the full Evidence-Weighted Federated Averaging algorithm.

**Algorithm 2:** Full EW-FedAvg with Adaptive Selection and Compression

**Input:** Initial model $\theta^0$, $K$ clients with datasets $\{\mathcal{D}_i\}$, local epochs $\tau$, step size $\eta$, rounds $T$, bandwidth budget $R_{\text{total}}$, cost threshold $c_{\min}$

**Output:** Trained global model $\theta^T$

```
// Phase 0: Variance estimation (run periodically)
for i = 1, ..., K do
    Compute σ̂_i^2 from pilot gradients at θ^0;
end
σ̂_min^2 ← min_i σ̂_i^2;
w_i ← σ̂_min^2 / σ̂_i^2 for each i;
for t = 0, 1, ..., T − 1 do
    // Step 1: Client selection (coalition formation)
    S_t ← {i : w_i > c_min} ;                    // exclude low-evidence clients
    W ← Σ_{i∈S_t} w_i;
    w̃_i ← w_i / W for each i ∈ S_t;
    Broadcast θ^t to all i ∈ S_t;
    // Step 2: Local training
    for each i ∈ S_t do                                          // in parallel
        θ_i ← θ^t;
        for k = 0, ..., τ − 1 do
            Sample mini-batch ξ ~ D_i;
            θ_i ← θ_i − η ∇F_i(θ_i; ξ);
        end
        Δ_i^t ← θ_i − θ^t;
    end
    // Step 3: Evidence-aware compression
    for each i ∈ S_t do
        Allocate bandwidth R_i ∝ w̃_i ‖Δ_i^t‖;
        k_i ← ⌊d · R_i / R_total⌋ ;                  // coordinates to send
        Δ̂_i^t ← TopK(Δ_i^t, k_i) ;                        // sparsification
    end
    // Step 4: Evidence-weighted aggregation
    θ^{t+1} ← θ^t + Σ_{i∈S_t} w̃_i Δ̂_i^t;
    // Step 5: Periodic variance re-estimation
    if t mod T_re-est = 0 then
        Update σ̂_i^2 from recent gradient statistics;
        Recompute w_i, w̃_i;
    end
end
return θ^T;
```

# 8 Discussion

## 8.1 Variance Estimation in Practice

The evidence weights $w_i = \sigma^2_{\min}/\sigma^2_i$ require estimates of client-level gradient variance. Two practical approaches:

(i) **Pilot phase**: at initialization, each client computes multiple gradient estimates at the same $\theta^0$ and reports the empirical variance to the server. Cost: $O(M)$ extra gradient computations per client, where $M$ is the number of pilot samples.

(ii) **Online estimation**: the server tracks the running variance of each client's updates $\Delta^t_i$ across rounds. After an initial burn-in period, these estimates stabilize and can be used for weighting. This requires no extra computation.

Both approaches preserve privacy: the server only sees gradient statistics, not raw data.

## 8.2 Connections to the Ω-Framework

The FL application demonstrates the generality of the Ω-framework's core principles:

(1) **Evidence weighting** ($w_i = V_{\min}/V_i$): the same weighting scheme improves convergence in both multi-agent RL and FL. The HM / AM improvement factor is a universal consequence of optimal inverse-variance weighting.

(2) **Self-knowledge bottleneck**: the insight that "you cannot communicate what you do not know" applies equally to RL agents communicating policies and FL clients communicating gradients. Bandwidth cannot substitute for data quality.

(3) **Coalition formation**: client selection in FL is structurally identical to coalition formation in cooperative games. The evidence weight serves as the natural inclusion criterion in both settings.

(4) **Blessing of dimensionality**: sparsified gradient communication exploits the same concentration phenomena that make high-dimensional policy gradient tractable.

## 8.3 Limitations

(i) **Variance estimation overhead**: the pilot phase adds communication rounds. Online estimation introduces a burn-in period during which weights may be inaccurate.

(ii) **Non-stationarity**: client data distributions may shift over time, requiring periodic re-estimation of $\sigma^2_i$.

(iii) **Privacy**: reporting $\hat{\sigma}^2_i$ leaks information about local data heterogeneity. Differential privacy mechanisms can be applied to the variance estimates at the cost of noisier weights.

(iv) **The evidence weight is not the only factor**: in practice, client reliability (uptime, network quality) and fairness considerations may override pure evidence-weighted selection.

# 9 Conclusion

We have shown that the Ω-framework's evidence-weighted multi-agent policy gradient transfers precisely to federated learning. The core mechanism — inverse-variance weighting via

$w_i = V_{\min}/V_i$ — produces a $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ variance improvement that is maximally beneficial precisely when data heterogeneity is worst. The self-knowledge bottleneck explains why bandwidth upgrades cannot fix the non-IID problem: clients with poor local data have fundamentally limited information about the global optimum, regardless of communication capacity. Evidence-weighted client selection provides a principled, automatic alternative to ad hoc selection heuristics.

The FL application validates the Ω-framework's claim to generality: its components (evidence weighting, lossy communication, coalition formation, blessing of dimensionality) are not artifacts of the multi-agent RL setting but universal principles of distributed learning under heterogeneous information quality.

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